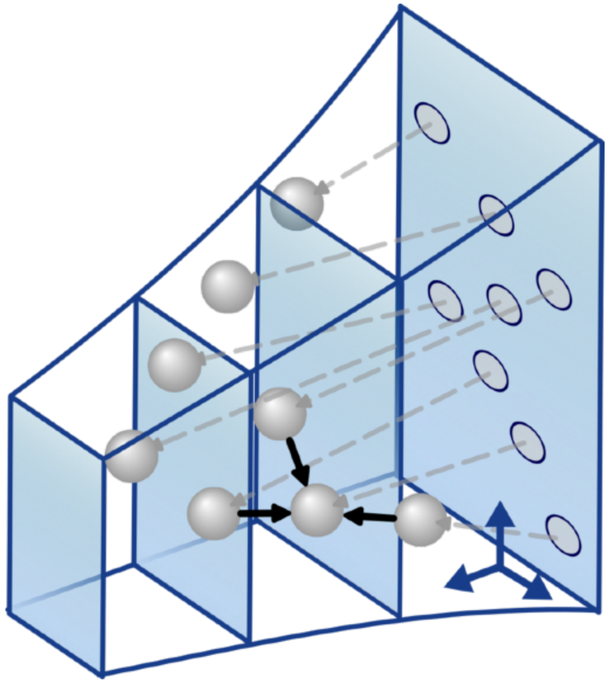
Why AdS?

- Convenient way to capture the **physically relevant part** of conformally invariant data.
- Conditioning on AdS proper distance naturally introduces **notion of locality** in scale.
- Idea of **"scale space"** has long history in computer vision — our use of AdS is a special version of scale space that also enforces special conformal transformation.
- This is an example of how ideas from a different field (i.e. string theory) allowed for a **genuinely novel solution** of an existing problem!



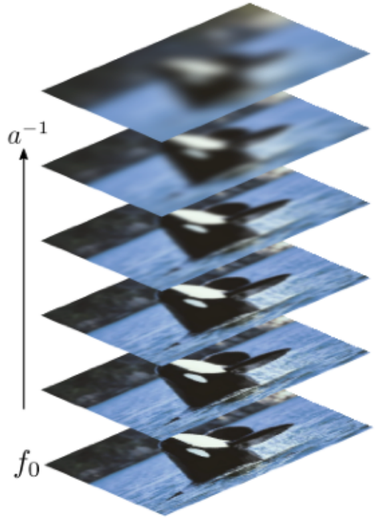


Figure 2: A Scale-space: For implementations we logarithmically discretize the scale-axis in a^{-1}.

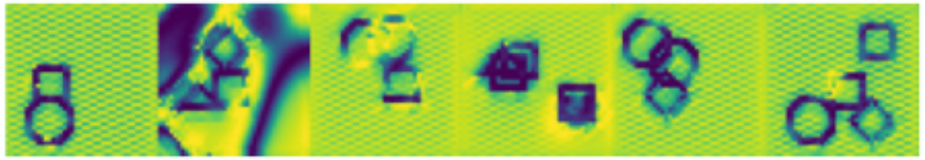
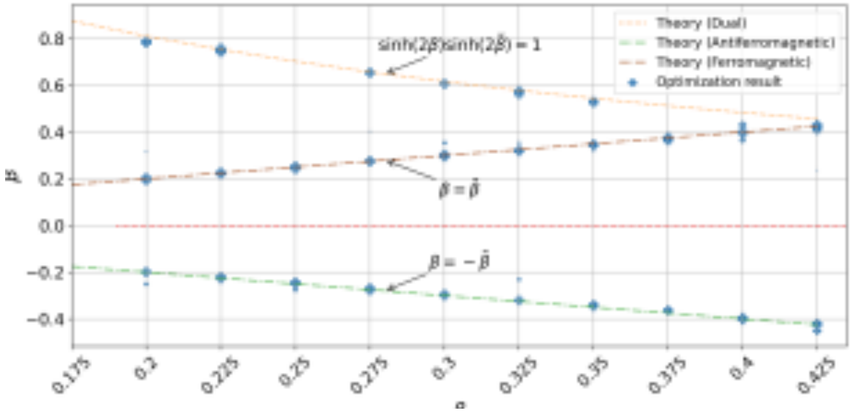
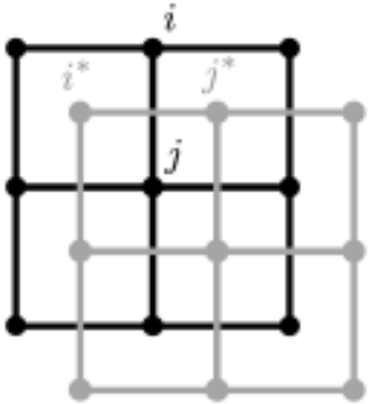
Worrall & Welling 2019

Future directions $t = \epsilon$

- **AdS-GNN+:** current model only uses scalar data, currently adding vector etc. data (results in better performance!)

- **Goldstone modes for deep learning:** understanding Kuramoto-type models for deep learning (Miyato et. al 2024) from the point of view of Goldstone modes and spontaneous symmetry breaking. Work in progress with M. Welling, T. Miyato, V. Vitelli.

- **Finding dualities with ML**: Training neural networks to find duality mappings in statistical physics (with P. Gupta, A. Ferrari: NeurIPS 2024 Workshop ML4PS, under review ICML).

- **Learning cluster algorithms from diffusion:** Cluster algorithms dramatically enhance MCMC simulation at critical points, but are only known for specific models: learn them in generalusing ideas from generative AI. Work in progress with R. Bondesan, A. Hossain.

- **Ask me afterwards!**